Ambalavanan M

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Executive Summary

A results-driven Frontend Developer and AWS Certified Cloud Practitioner with hands-on experience in Java, Python, C++, and modern Web Technologies. Skilled in building scalable cloud architectures on AWS and developing end-to-end full-stack applications integrating Machine Learning and Blockchain. Passionate about solving complex engineering challenges and delivering innovative, production-ready software solutions.

Technical Skills

- Programming Languages: Java, Python, C++, C, Solidity
- Web Technologies: HTML5, CSS3, JavaScript (ES6+)
- Cloud Platforms: AWS S3, AWS Lambda, AWS API Gateway, AWS CloudFront, AWS SageMaker
- Frameworks & Libraries: React.js
- Databases & Concepts: SQL, DBMS, Data Structures & Algorithms (DSA), Object-Oriented Programming (OOP)
- Tools & Version Control: Git, GitHub

Soft Skills

- Problem Solving & Analytical Thinking
- Communication & Teamwork
- Time Management & Adaptability
- Self-Learning & Continuous Improvement

Projects

Loan Approval Prediction Portal 2026

Role: Full-Stack Developer / ML Engineer

Responsibilities:

- Designed and deployed an end-to-end ML web application to predict loan approval status using historical financial data.
- Trained a Linear Regression model on AWS SageMaker with structured financial datasets.
- Architected a serverless backend using AWS Lambda and API Gateway, reducing infrastructure overhead and enabling auto-scaling.
- Built a React.js frontend interface enabling real-time loan eligibility assessment, integrating directly with AWS cloud services via REST API.
- Stored and managed datasets and model artifacts on AWS S3, ensuring secure and reliable data access across all application components.
- Coordinated end-to-end deployment pipeline from data preprocessing to cloud-hosted production environment.

Technologies Used: Python, Machine Learning, AWS SageMaker, AWS Lambda, AWS API Gateway, AWS S3, React.js

E-Voting System Using Blockchain 2026

Role: Blockchain Developer / Frontend Developer

Responsibilities:

- Developed a decentralized e-voting application using Solidity smart contracts on the Ethereum blockchain, ensuring tamper-resistant digital election workflows.
- Implemented secure voter authentication and immutable vote recording to eliminate fraud and maintain auditability.
- Built a responsive web frontend using HTML, CSS, and JavaScript to interact with deployed smart contracts through the browser.
- Ensured end-to-end transparency in the election process through blockchain-based audit trails.
- Tested and validated smart contract logic for correctness, security vulnerabilities, and edge cases.

Technologies Used: Solidity, Blockchain (Ethereum), HTML, CSS, JavaScript

Crisis Response Management System 2026

Role: Frontend Developer

Responsibilities:

- Built a responsive web application to streamline emergency coordination, real-time resource tracking, and crisis communication among first responders.
- Designed an intuitive JavaScript-driven interface for incident reporting, resource allocation, and status updates.
- Implemented dynamic UI components to support multi-user workflows for crisis management teams.
- Optimized the interface for quick load times and accessibility to support time-sensitive emergency scenarios.
- Collaborated on requirements gathering and feature design to align deliverables with real-world crisis response needs.

Technologies Used: HTML, CSS, JavaScript

Education

B.Sc. Computer Science Expected Graduation: 2027

VIT University, Vellore

Certifications & Accomplishments

- AWS Cloud Foundations – Amazon Web Services (AWS)
- AWS Certified Cloud Practitioner – Amazon Web Services (AWS)
- Academic Excellence – CGPA: 9.31 / 10, B.Sc. Computer Science, VIT University